Chapter 2.

Natural Language Processing

Learning objectives

- Understand what NLP is and its real-world applications
- Explain why text is challenging for computers
- Perform text preprocessing (cleaning, tokenization, normalization)
- Implement keyword search (simple and TF-based)
- Start to understand sparse vs dense vector
- Implement similarity-based search using TF-IDF
- Cluster documents using machine learning (in order to visualize similarity)

What is Natural Language Processing?

NLP = Teaching computers to **understand**, **interpret**, and **generate human language**

🔍 **Search engines**: Understanding what you really mean when you search, and finding it

📧 **Email filtering**: Identifying spam vs important messages

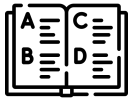
💬 **Chatbots**: Understanding and replying to customer questions

📊 **Document analysis**: Organizing and finding information in large document collections

🎯 **Recommendation systems**: Finding similar content (movies, products, jobs)

… so much more

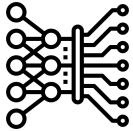
NLP Use cases



Representative models

Retrievals

Text Classification



Generative models

Text translation

Text generation

Let’s start with a demo

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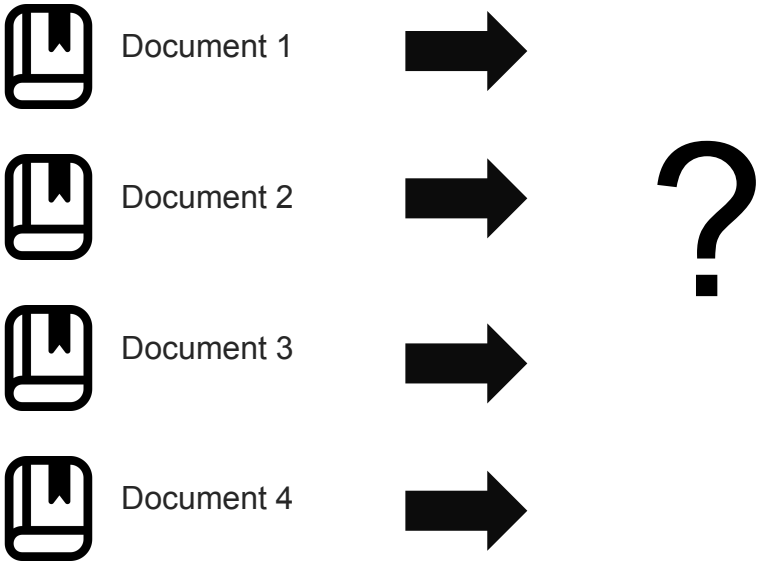
You're building a system to help recruiters search through thousands of CVs/resumes.

The Challenge: A recruiter searches for **"Python developer with machine learning experience"** but needs to find candidates who wrote:

- "Proficient in Python and ML"
- "Experienced in data science using Python"
- "Strong background in artificial intelligence and Python programming"

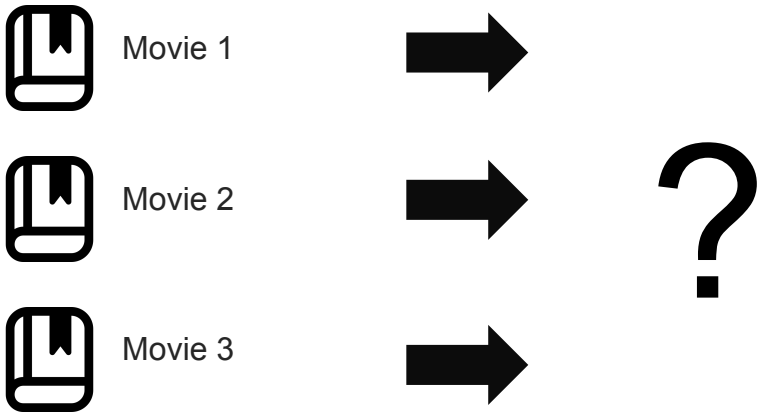
Simple keyword matching fails! The same meaning is expressed differently.

NLP = Teaching computers to **understand**, **interpret**, and **generate human language**


Document 1
Document 2
Document 3
Document 4
?

I want to look for movies about space adventures

Will I enjoy this movie?



Syntactic models

Syntactic models focus on the structure and form of language, analyzing how words and symbols are arranged according to grammatical or formal rules.

Semantic models

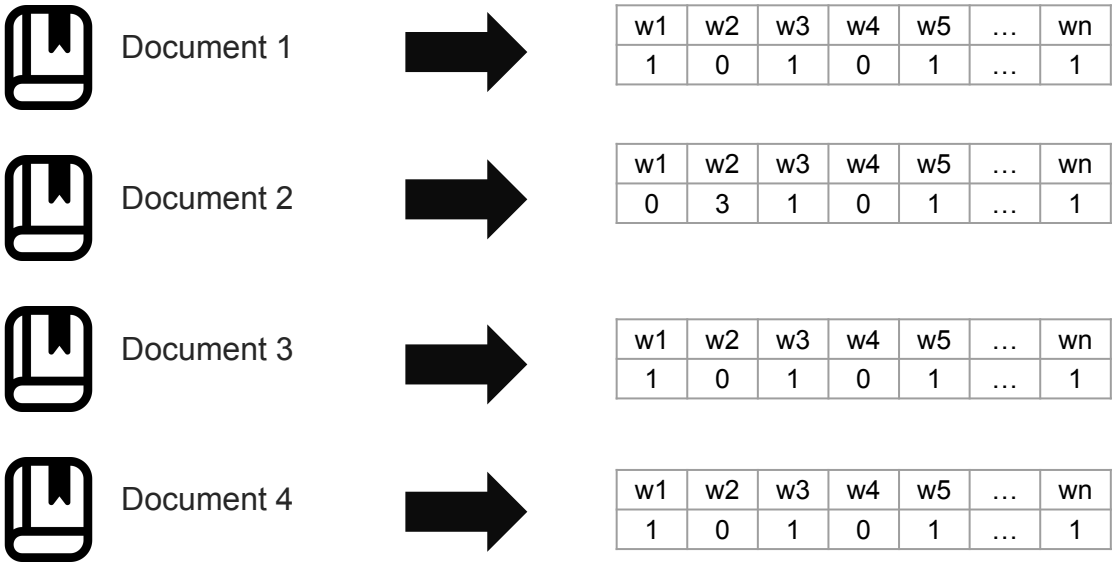
Semantic models capture meaning and relationships between concepts, interpreting the content and context of language beyond its surface structure.

Syntactic models

"Why Is The Rum Always Gone?"

Why	is	the	rum	always	gone	Other words
1	1	1	1	1	1	…

Syntactic models



Bag of Words

The **Bag-of-Words (BoW) Model** is a feature extraction technique in NLP that converts text into **numerical vectors** by simply **counting the occurrence of each word** in a document. This vector representation allows machine learning algorithms to process text data. Its main limitation is that it **completely ignores word order and grammatical structure**.

Syntactic models

Simple but they “understand” only syntax.

Syntactic models

Movie 1: An amazing creature **is running** from a hunter.
['an': 1, 'amazing': 1, 'creature': 1, 'is': 1, 'running': 1, 'from': 1, 'a': 1, 'hunter': 1, '.': 1]

Movie 2: The hero **ran** through the city to **escape**.
['the': 2, 'hero': 1, 'ran': 1, 'through': 1, 'city': 1, 'to': 1, 'escape': 1, '.': 1]

Movie 3: Running away, the villain **escapes** the **runners!**
['running': 1, 'away': 1, ',': 1, 'the': 2, 'villain': 1, 'escapes': 1, 'runners!': 1, '!': 1]

Movie 4: A fun, but slightly **boring**, fantasy film.
['a': 1, 'fun': 1, ',': 1, 'but': 1, 'slightly': 1, 'boring': 1, 'fantasy': 1, 'film': 1, '.': 1]

Chapter 2.

Natural Language Processing

Formally speaking we could define search as:

$$F(q,c) = y$$

I.e a function that given a **query** q and a **corpus** c we get a set of **relevant** results y

Then, building a good search system means **optimizing** each part:

- crafting the right **query representation** q,
- structuring or *embedding* the **corpus** c effectively,
- and designing or learning the best **retrieval function** F

Query: How the user expresses intent, can we make it **clearer**, **richer**, or **smarter**?

Corpus: How the information is **stored** and **represented**, can we make it easier to match, faster to search, or richer in features?

Function: How we **match** and **rank**, can we use better algorithms, learning signals, or hybrid strategies?

Query (q)	Corpus (c)	Retrieval Function (F)
Text representation / embeddings	Raw vs. processed content	Matching strategy (keywords, vectors)
Query expansion / synonyms	Indexing (inverted, vector)	Ranking (TF-IDF, BM25, learned)
Contextualization (history, intent)	Preprocessing / cleaning	Learning & adaptation (ML, feedback loops)
Structured queries / filters	Metadata / structure	Hybrid approaches (lexical + semantic)

Chapter 2.

Natural Language Processing

Syntactic models

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Syntactic models

Vocabulary

all unique words in a corpus; each word becomes a feature in syntactic models

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An amazing creature is running from a hunter

[1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]

Stemming is the process of reducing a word to its base or root form by removing prefixes or suffixes — often using simple, rule-based heuristics (e.g., *"running"* → *"run"*, *"flies"* → *"fli"*).

Lemmatization is the process of reducing a word to its canonical or dictionary form (*lemma*) using linguistic knowledge, taking into account the word's meaning and part of speech (e.g., *"better"* → *"good"*, *"running"* → *"run"*).

Syntactic models

But are all those **tokens** necessary?

all unique ~~words~~ tokens in a corpus; each word becomes a feature in syntactic models

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Syntactic models

But, are all those **tokens** necessary?

Not really. There are two standard ways to handle this in syntactic models:

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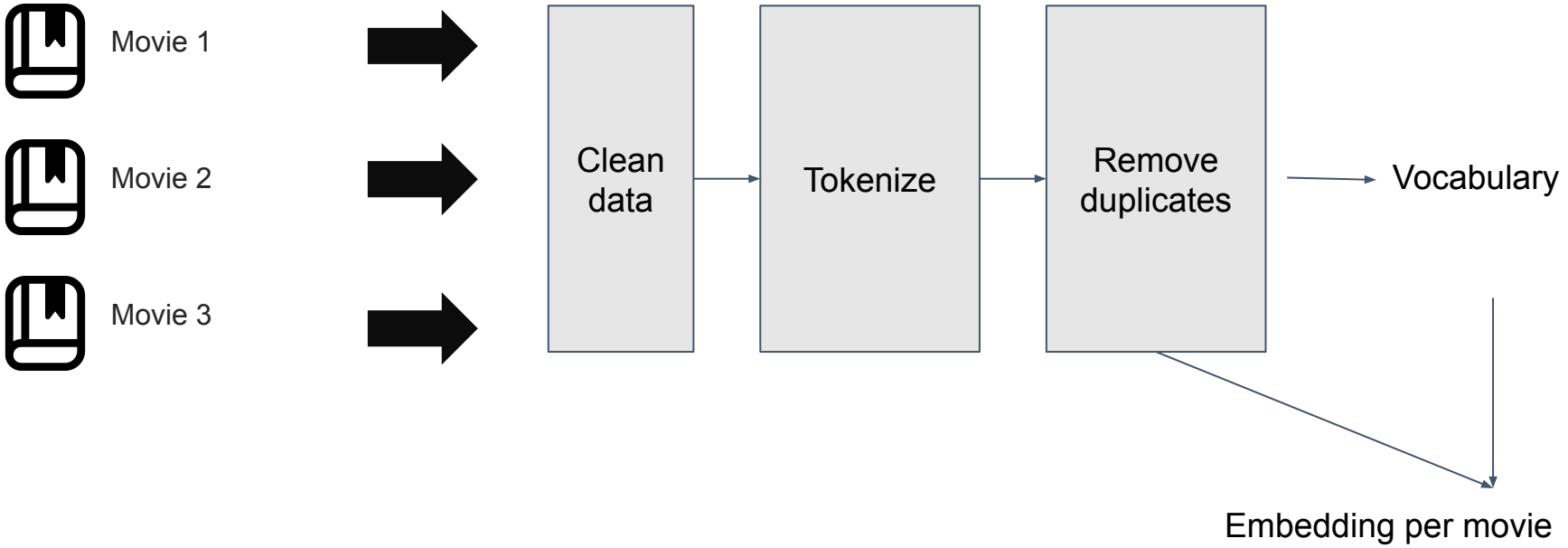
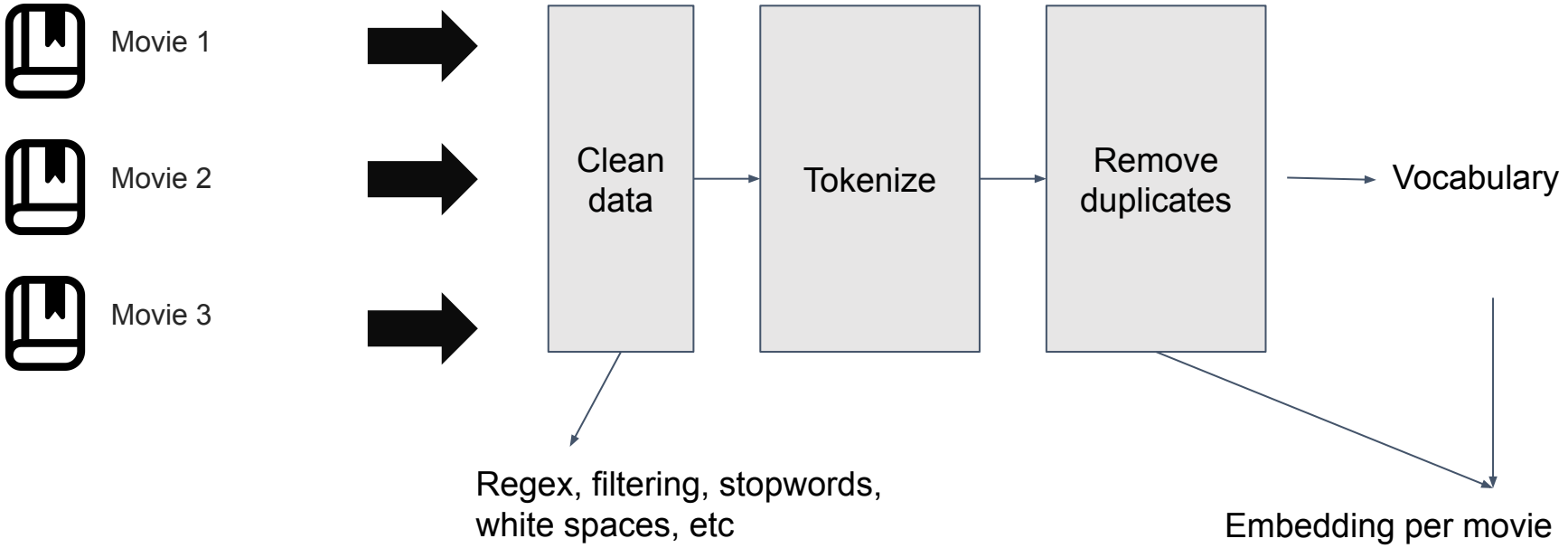
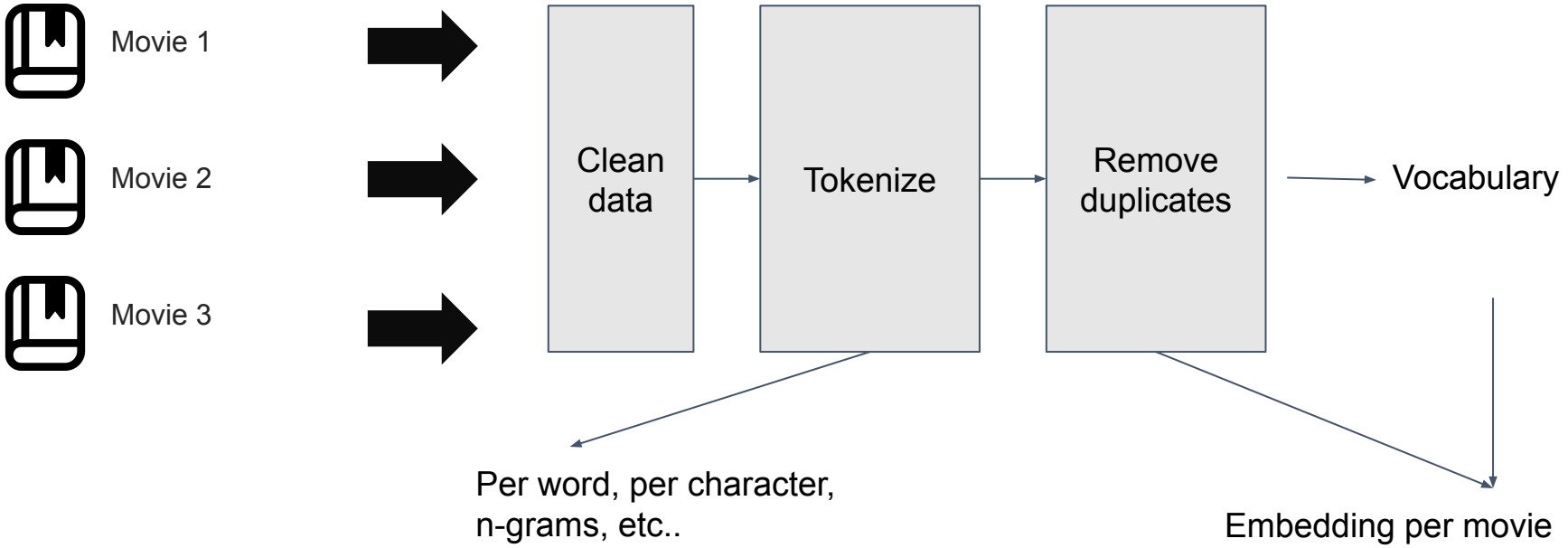

Movie 1
Movie 2
Movie 3
Clean data
Tokenize
Remove duplicates
Vocabulary
Embedding per movie


Movie 1
Movie 2
Movie 3
Clean data
Tokenize
Remove duplicates
Vocabulary
Regex, filtering, stopwords, white spaces, etc
Embedding per movie


Movie 1
Movie 2
Movie 3
Clean data
Tokenize
Remove duplicates
Vocabulary
Per word, per character, n-grams, etc..
Embedding per movie

Natural Language Processing

What are embeddings ?

Embeddings are vectors that represent real-world objects, like **words**, images, or videos, in a form that **computer algorithms** like machine learning models can easily process.

	Feature #1	Feature #2	Feature #3	Feature #4	...
Document #1	1	0	1	0	0
Document #2	0	0	2	0	0
Document #3	0	1	0	0	1
Document #4	0	0	0	1	0
⋮	0	0	1	0	0

DTM

TL;DR it's just a bunch of numbers with meaning for computers

[Let, ‘s, do ,stuff]

Chapter 2.

Natural Language Processing


LANGUAGE
MODEL
TEXT
VOCABLARY
SENTENCE
MEANING

Are all documents, movies,.. the same?

Is just the frequency of the word enough?

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V = {cats, this, is, about, dogs, like, to, bark, jump, around}

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A possible solution?

TF-IDF = Term Frequency x Inverse Document Frequency

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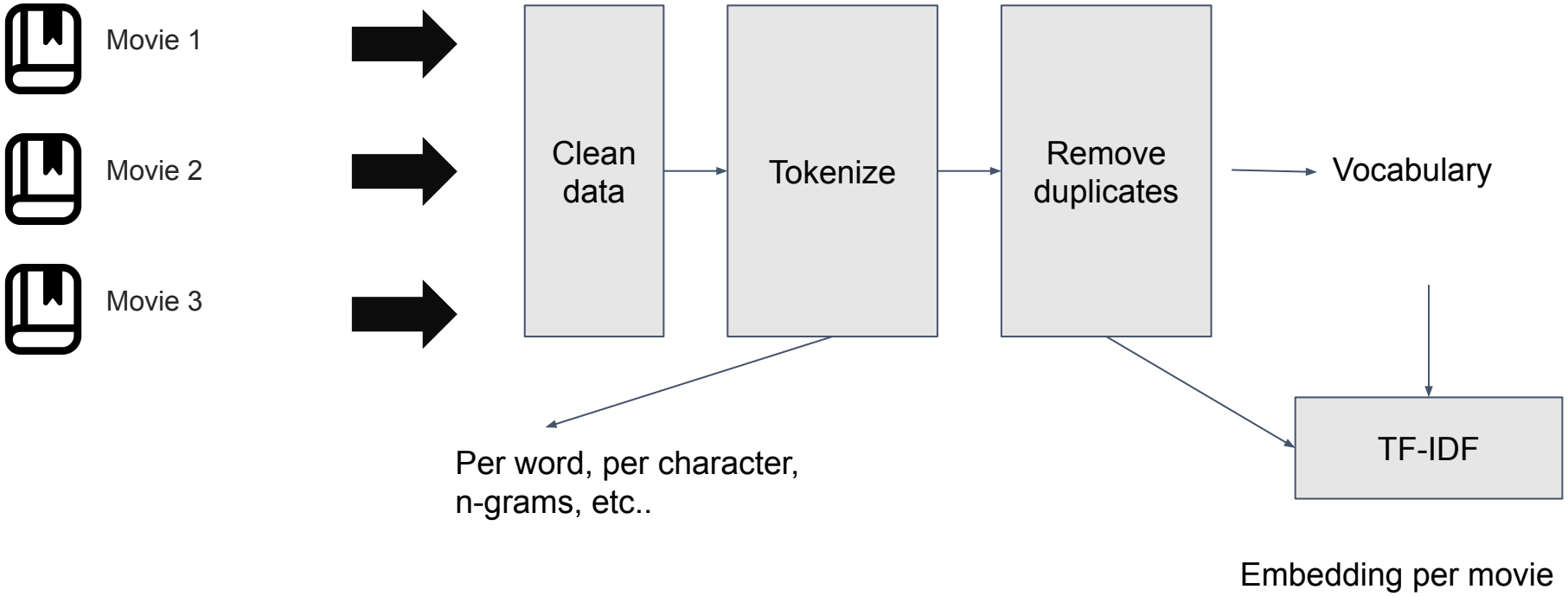
No, the length of the document matters.
Is just the frequency of the word enough?
No. The loudest doesn't always win.

A possible solution?

TF-IDF = Term Frequency x Inverse Document Frequency

Term Frequency = Bag of Words (the frequency of each word in the vocabulary)

Inverse Document Frequency = How unique a word is across **all** documents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TF-IDF
Per word, per character, n-grams, etc..
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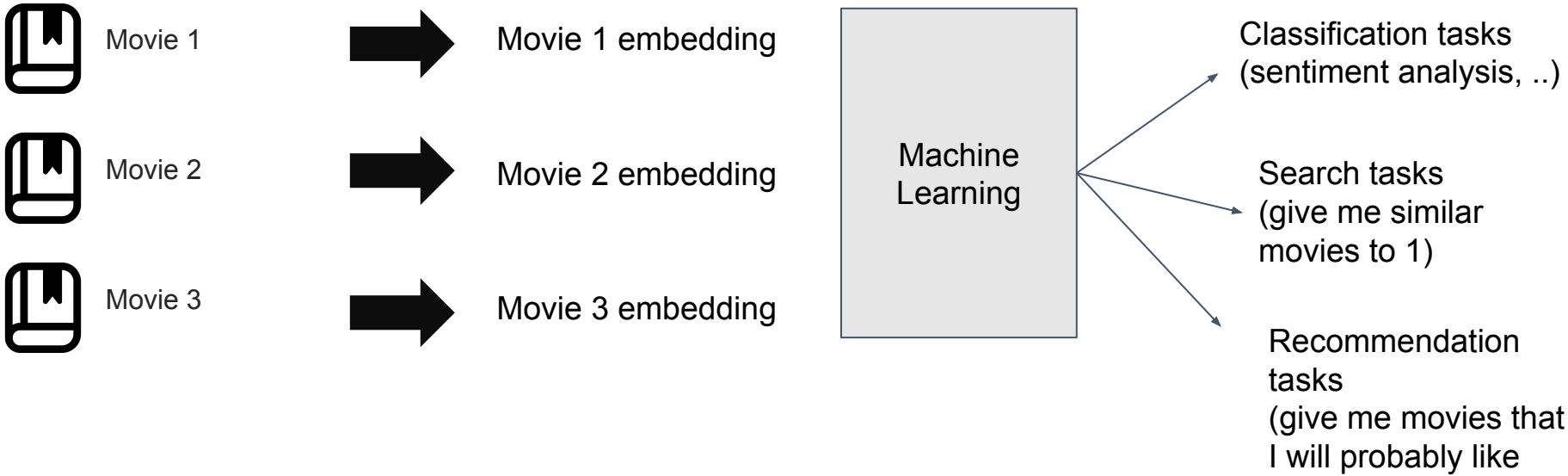
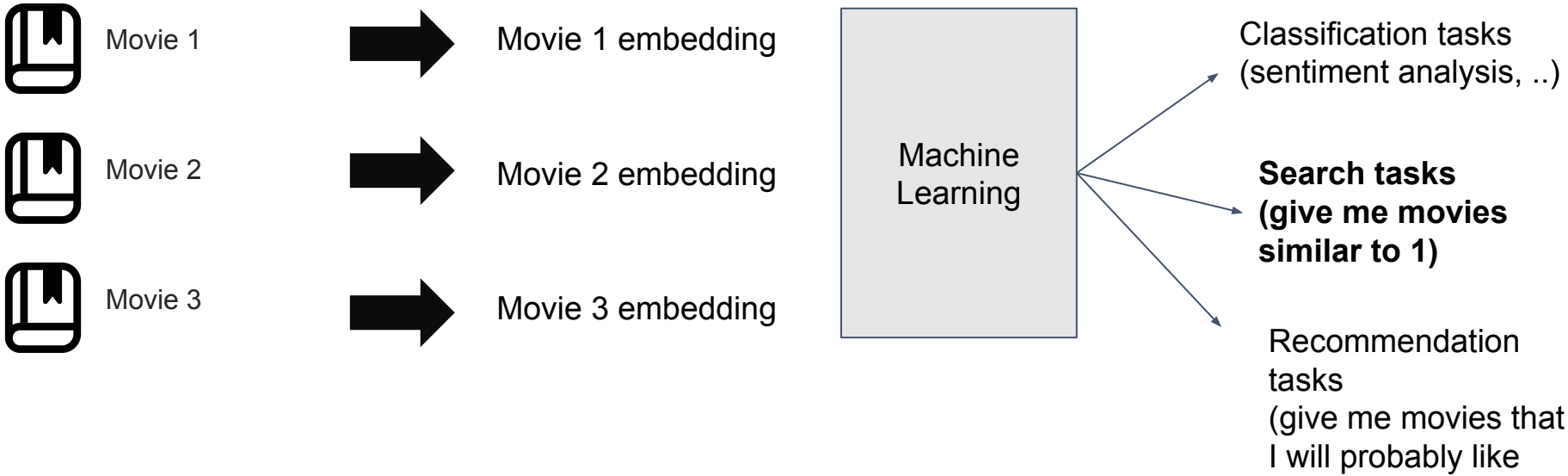

Movie 1
Movie 2
Movie 3
Movie 1 embedding
Movie 2 embedding
Movie 3 embedding
Machine Learning
Classification tasks (sentiment analysis, ..)
Search tasks (give me similar movies to 1)
Recommendation tasks (give me movies that I will probably like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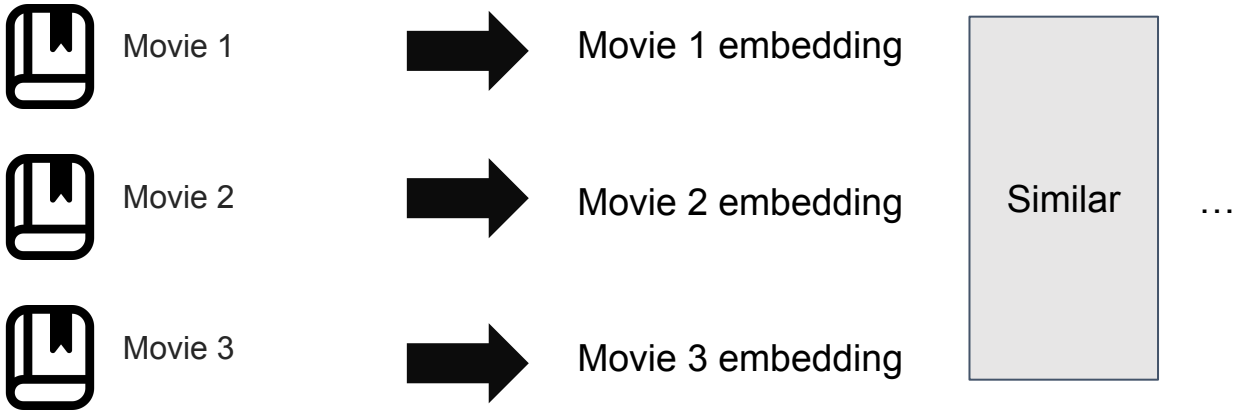
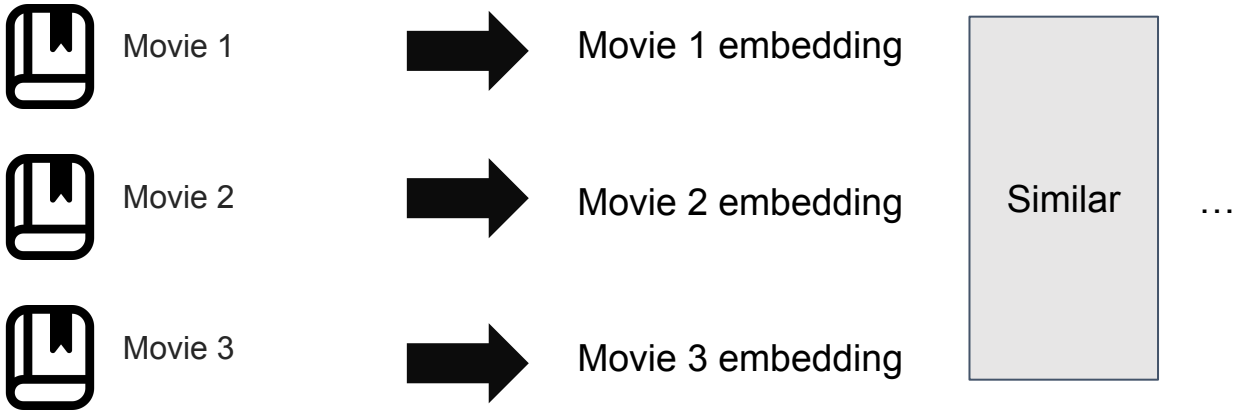

Movie 1
Movie 2
Movie 3
Movie 1 embedding
Movie 2 embedding
Movie 3 embedding
Similar
…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Embedding = Vector

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DTM

TL;DR it's just a bunch of numbers with meaning for computers

Vector Space Model

Treat each document as a **point in space**, where mathematical tools can be used to measure distances and relationships.

Recall that the TF-IDF process converts a document into a list of weighted scores, one for every unique word in the vocabulary.

Definition: In a **Vector Space Model (VSM)**, a document is represented as a **vector.**

- Each **dimension** of the space corresponds to a unique word in the corpus vocabulary.
- The **value** along each dimension is the calculated TF-IDF weight for that word in the document.

{cats, this, is, ….. around}. This means our VSM is a **10-dimensional space**.

Vector Space Model

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Some similarities for exact nearest neighbor:

- Cosine Similarity
- Euclidean Similarity
- Jaccard Similarity

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- Euclidean Similarity
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$$\text{cosine similarity} = S_C(A,B) := \cos(\theta) = \frac{\mathbf{A}\cdot\mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\cdot\sqrt{\sum_{i=1}^{n} B_i^2}},$$

Cosine similarity measures **how similar two documents** are by comparing the angle between their vector representations in a **high-dimensional space**.
Each document is first converted into an **embedding**, a numerical vector capturing its *semantic meaning*.
The cosine similarity score, **which ranges from -1 to 1**, indicates **how closely the documents align in meaning**: a score near 1 means they are very similar, while a score near 0 means they are unrelated.
In document-to-document search, this **metric** allows systems to **find** and **rank** documents with content that is **semantically close**, regardless of differences in length or word choice, making it ideal for **clustering**, **recommendation**, and **information retrieval tasks**.

Note: Although cosine similarity theoretically ranges from –1 to +1, real-world document embeddings rarely produce negative values—most unrelated texts fall near 0.

Vector Space Model

Treat each document as a **point in space**, where mathematical tools can be used to **measure distances and relationships.**

P.s in the real world with millions of documents,where speed and costs matter, we tend to use Approximate Nearest Neighbors (ANN) algorithms instead of Exact Nearest Neighbors, one of the most famous ones is the Hierarchical Navigable Small Worlds.

[Let, ‘s, do ,stuff, again]